

Technology Stack Specification

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML5, Vanilla CSS3 (Custom Glassmorphism), Vanilla JavaScript (ES6+), Chart.js library	HTML5 and CSS3 deliver semantic layouts and fully responsive design using Grid and Flexbox. Custom CSS glassmorphism (backdrop-blur) yields a premium modern visual look. Vanilla JS avoids package overhead. Chart.js compiles interactive graphs comparing model metrics.
2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python (v3.12), Flask Framework (v3.1.3), Gunicorn WSGI Server	Python is the standard language for machine learning modeling and preprocessor integrations. Flask is a lightweight micro-framework that runs scoring REST endpoints (/api/predict) with negligible latency and high processing efficiency.

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3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	Model Storage serialization (pickle binaries: model.pkl, preprocessor.pkl), Session-persistent LocalStorage	Pickled serialization of model parameters and StandardScaler objects bypasses heavy SQL database query structures, loading in <10ms. Session-persistent LocalStorage handles evaluation history tables directly in the browser.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	IBM Watson Machine Learning (WML), IBM Cloud Object Storage	Watson ML allows registering models and deploying them as active cloud scoring endpoints. Cloud Object Storage securely caches the pickled model packages, facilitating global API access.
5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	Git versioning, GitHub code hosting	Git tracks code changes during staging and ensures clean rollbacks. GitHub acts as the remote hosting repository, safeguarding collaboration logs.

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6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Scikit-Learn (sklearn), Pandas, NumPy, FontAwesome Icons, Google Fonts (Outfit, Inter)	Scikit-learn provides ML estimators, evaluators, and preprocessors. Pandas and NumPy handle data matrix ingestion. FontAwesome and Google Fonts ensure custom, high-definition icons and modern typography styles.